Exp1

Introduction to numpy for Data Analysis

Student	Maths	Physics	Chemistry	English
S_1	85	78	92	88
S_2	76	81	85	79
S_3	90	95	89	93
S_4	65	72	70	68
S_5	88	84	91	90

a) Display the marks of all students.

```
import numpy as np
mark = np.array([
    [85, 78, 92, 88],
    [76, 81, 85, 79],
    [90, 95, 89, 93],
    [65, 72, 70, 68],
    [88, 84, 91, 90]
])
print(mark)
```

b) Find the following:

i) shape of the array

```
print(mark.shape)
```

ii) Number of dimensions

iii) Size of Array

print (mark. size)

iv) Data TYPE

print (mark. dtype)

c) Display :

i) Marks of Students

print (mark[2])

ii) Marks scored in chemistry by all Students

print (mark[:,2])

iii) Mark of Student 5 in English

print (mark [4,3])

d) Calculate

i) Total marks obtained by each student

tot = np. sum (mark, axis=1)

print (tot)

ii) Average marks of each students

avg = np. mean (mark, axis=1)

print (avg)

e) Calculate

i) Average marks in each subject

```
sub_avg = np.mean (mark, axis=0)
print (sub_avg)
```

ii) Identify the subject with the highest average marks

```
sub = ["Maths", "Physics", "Chemistry", "English"]
highest = np.argmax (sub_avg)
print (sub[highest])
```

f) Find:

i) Highest marks in the dataset

```
print (np.max (mark))
```

ii) Lowest mark in the dataset

```
print (np.min (mark))
```

g) Display all students who scored more than 90 marks in any subject.

```
result = mark[np.any (mark > 90, axis=1)]
print (result)
```

h) Add 5 grace marks to every students English score and display the updated marks

```
upt_mark = mark.copy()
upt_mark [:,3] += 5
print (upt_mark)
```

I) Reshape the marks array into a 2×10 matrix and display the result

```
new_mark = mark.reshape(2,10)
print (new_mark)
```

J) Create a 5×5 Identity Matrix using numpy

```
Identity = np.eye(5)
print (Identity)
```

